SpO2 and Body Temperature Measurement Device

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Abstract

- The progress of science and technology must be able to help resolve human medical problems for human survival. Integrated measurement of oxygen saturation in the blood, heart rate and temperature of the human body have been made into one tool whose measurement results are displayed directly on the LCD display on a measuring instrument and an Android Smartphone. This tool uses Arduino Nano as its Microcontroller, Bluetooth as wireless serial communication with Android smartphone, Android Application Made with MIT App Inventor, MAX30102 sensor is used to measure blood oxygen saturation and heart rate, and GY-906-BCC sensor to measure body temperature. Testing Measurements that have been made are calibrated with Industrial standard measuring instruments (Oximeters) and digital thermometers. The result is that the accuracy of the tool reaches 99.62% for blood oxygen saturation measurements, 97.55% for heart rate, and 99.62% for body temperature when compared with industry standard devices. Android applications have been successfully installed on 5 types of smartphones with different brands and specifications. The Bluetooth range of a measuring device for a smartphone in maximum data transfer is 29 meters without a hitch and a maximum of 21.5 meters with obstacles.

Introduction

- What is SpO2?:-SpO2 is the saturation percentage of oxygen in the blood. SpO2 is measured at the periphery, usually a finger, and is one measure of the health of the cardiovascular and respiratory systems.
- SPO2 is measured using a sensor, usually attached to the patient’s finger.
- In our project we will be using the Max 30102 Heart Rate sensor and GY-906-BCC Non-Contact Infrared Temperature Sensor Module. To determine the body temperature and the SpO2 levels of the body.
- Pulse oximetry is a non-invasive method for measuring the percentage rate of hemoglobin saturation. This method uses the different wavelengths of red light (660 nm) and infrared light (940 nm) captured by the detection sensors after passing through the veins and capillaries at the tip of the index finger

Methodology

This project can detect the SpO2 levels and body temperature at the same time, any value lower than the set value will light up the red LED, otherwise the green LED would light up.

We will be using an Arduino Uno to program the device and a 20X4 LCD Display to show the detected SpO2 levels and the body temperature.

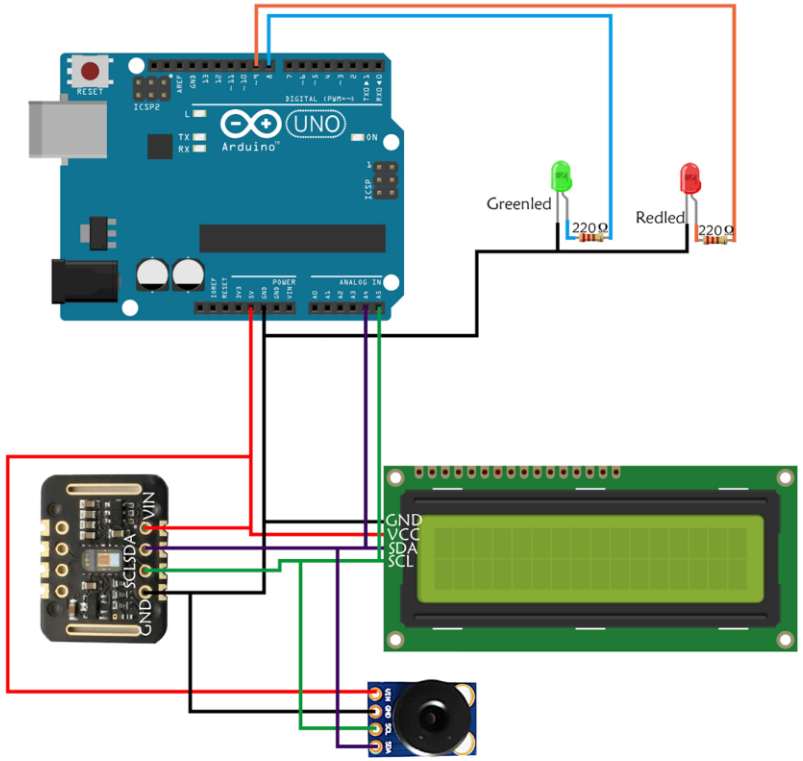
Hardware Components:

1. Arduino UNO
2. 20x4 I2C LCD
3. Max30102 Heart Rate Sensor
4. Two LEDs
5. 220 ohm resistors x2
6. GY-906-BC Non-Contact Infrared Temperature Sensor Module
7. Breadboard

- Software used:

- Arduino IDE

Circuit Diagram



Algorithm/Code

```
MAX30105 particleSensor;
LiquidCrystal_I2C lcd(0x27,20,4);
//#define MAX30105 //if you have Sparkfun's MAX30105 breakout board , try #define MAX30105
Adafruit_MLX90614 mlx = Adafruit_MLX90614();

double avered = 0; double aveir = 0;
double sumirrms = 0;
double sumredrms = 0;
int i = 0;
int Num = 100;//calculate SpO2 by this sampling interval
int Temperature;
int temp;
float ESpO2;//initial value of estimated SpO2
float ESpO2_ROM;
double FSpO2 = 0.7; //filter factor for estimated SpO2
double frate = 0.95; //low pass filter for IR/red LED value to eliminate AC component
#define TIMETOBOOT 3000 // wait for this time(msec) to output SpO2
#define SCALE 88.0 //adjust to display heart beat and SpO2 in the same scale
#define SAMPLING 5 //if you want to see heart beat more precisely , set SAMPLING to 1
#define FINGER_ON 30000 // if red signal is lower than this , it indicates your finger is not on the sens
#define USEFIFO
#define Greenled 8
#define Redled 9
void setup()
{
  Serial.begin(115200);

  lcd.init();
  lcd.backlight();
  lcd.setCursor(3,1);
```

Algorithm/Code

```
lcd.print("Running......");
delay(3000);
lcd.clear();
ESpO2 = readEEPROM();
Temperature = EEPROM.read(6);
pinMode(Greenled,OUTPUT);
pinMode(Redled,OUTPUT);
digitalWrite(Greenled,LOW);
digitalWrite(Redled,LOW);
// Initialize sensor
while (!particleSensor.begin(Wire, I2C_SPEED_FAST)) //Use default I2C port, 400kHz speed
{
  Serial.println("MAX30102 was not found. Please check wiring/power/solder jumper at MH-E
  //while (1);
}

//Setup to sense a nice looking saw tooth on the plotter
byte ledBrightness = 0x7F; //Options: 0=Off to 255=50mA
byte sampleAverage = 4; //Options: 1, 2, 4, 8, 16, 32
byte ledMode = 2; //Options: 1 = Red only, 2 = Red + IR, 3 = Red + IR + Green
//Options: 1 = IR only, 2 = Red + IR on MH-ET LIVE MAX30102 board
int sampleRate = 200; //Options: 50, 100, 200, 400, 800, 1000, 1600, 3200
int pulseWidth = 411; //Options: 69, 118, 215, 411
int adcRange = 16384; //Options: 2048, 4096, 8192, 16384
// Set up the wanted parameters
particleSensor.setup(ledBrightness, sampleAverage, ledMode, sampleRate, pulseWidth, adcRa

particleSensor.enableDIETEMPRDY();
mlx.begin();


void loop()


uint32_t ir, red , green;
```

Algorithm/Code

```
  uint32_t ir, red , green;
  double fred, fir;
  double SpO2 = 0; //raw SpO2 before low pass filtered

#ifdef USEFIFO
  particleSensor.check(); //Check the sensor, read up to 3 samples

  while (particleSensor.available()) {//do we have new data
#ifdef MAX30105
   red = particleSensor.getFIFORed(); //Sparkfun's MAX30105
    ir = particleSensor.getFIFOIR();  //Sparkfun's MAX30105
#else
    red = particleSensor.getFIFOIR(); //why getFOFOIR output Red data by MAX30102 on MH-ET LIVE breakout board
    ir = particleSensor.getFIFORed(); //why getFIFORed output IR data by MAX30102 on MH-ET LIVE breakout board
#endif



    i++;
    fred = (double)red;
    fir = (double)ir;
    avered = avered * frate + (double)red * (1.0 - frate);//average red level by low pass filter
    aveir = aveir * frate + (double)ir * (1.0 - frate); //average IR level by low pass filter
    sumredrms += (fred - avered) * (fred - avered); //square sum of alternate component of red level
    sumirrms += (fir - aveir) * (fir - aveir);//square sum of alternate component of IR level
    if ((i % SAMPLING) == 0) {//slow down graph plotting speed for arduino Serial plotter by thin out
      if ( millis() > TIMETOBOOT) {
        float ir_forGraph = (2.0 * fir - aveir) / aveir * SCALE;
        float red_forGraph = (2.0 * fred - avered) / avered * SCALE;
        //trancation for Serial plotter's autoscaling
        if ( ir_forGraph > 100.0) ir_forGraph = 100.0;
        if ( ir_forGraph < 80.0) ir_forGraph = 80.0;
        if ( red_forGraph > 100.0 ) red_forGraph = 100.0;
        if ( red_forGraph < 80.0 ) red_forGraph = 80.0;
```

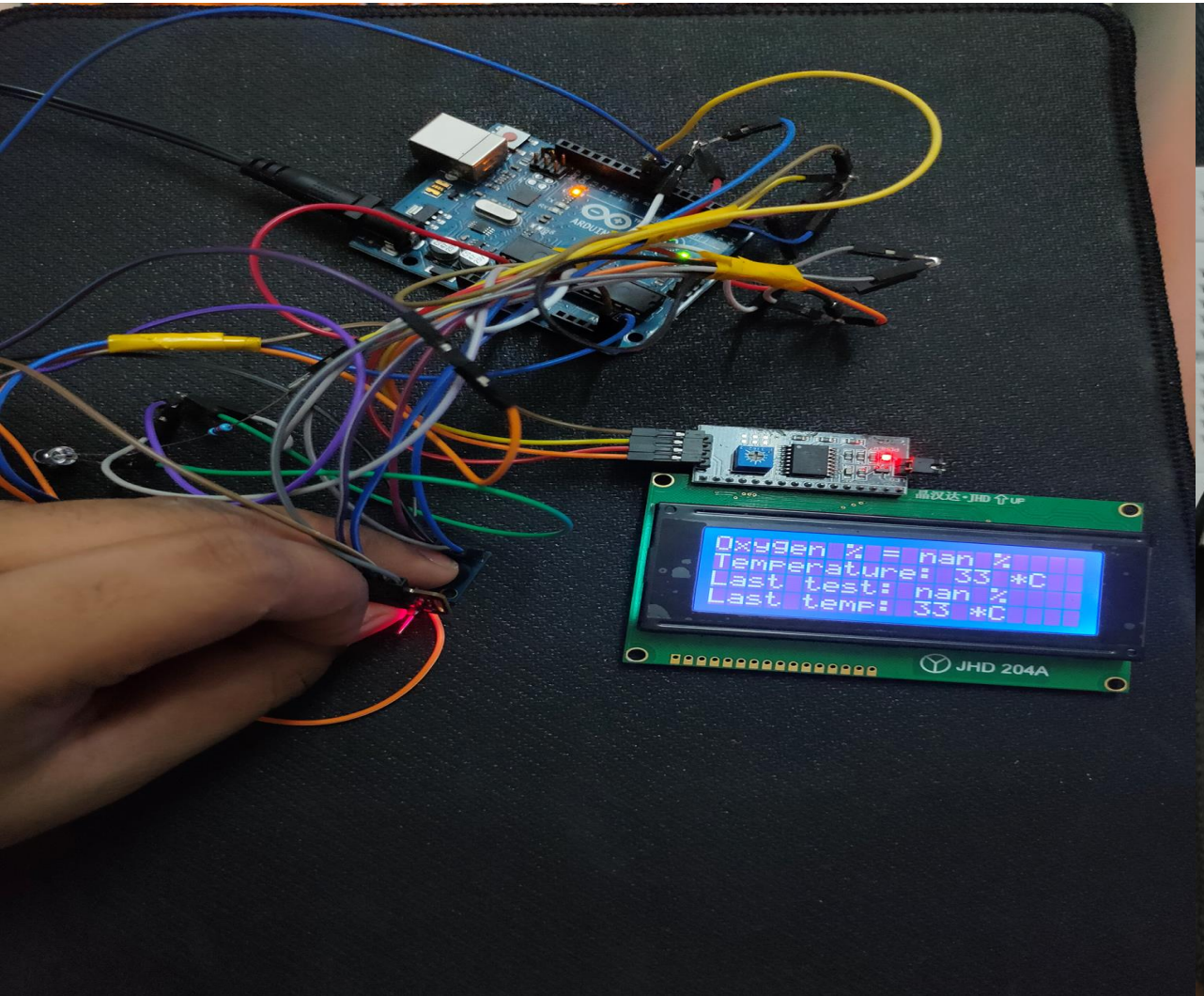
Algorithm/Code

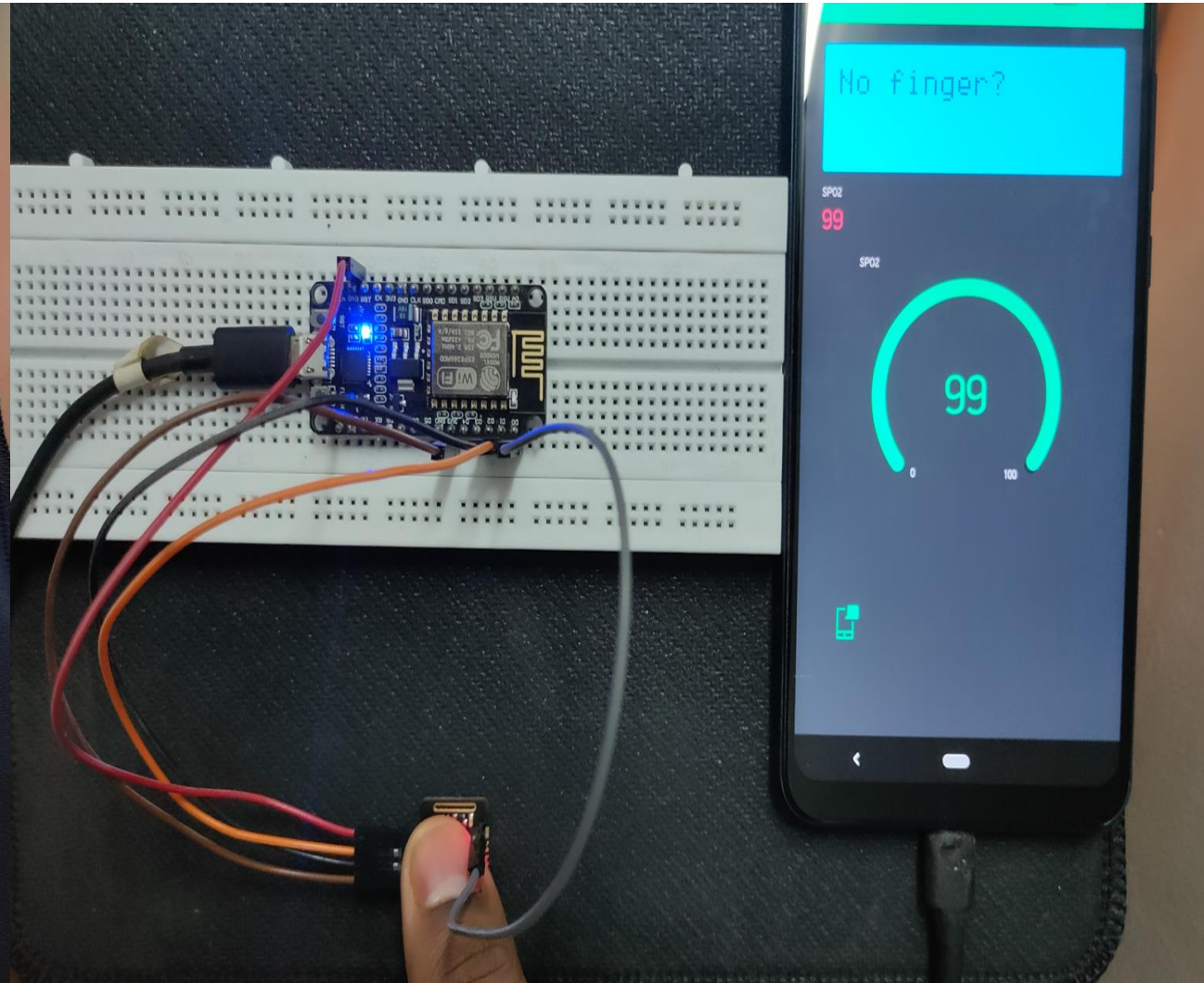
```
//        Serial.print(red); Serial.print(","); Serial.print(ir);Serial.print
float temperature = particleSensor.readTemperatureF();

if (ir < FINGER_ON){
temp = Temperature;
EEPROM.write(6,temp);
ESpO2_ROM = ESpO2;
writeEEPROM(&ESpO2_ROM);
lcd.setCursor(0,2);
lcd.print("Last test: ");
lcd.setCursor(11,2);
lcd.print(ESpO2_ROM);
lcd.print(" ");
lcd.print("% ");

lcd.setCursor(0,3);
lcd.print("Last temp: ");
lcd.setCursor(11,3);
lcd.print(temp);
lcd.print(" ");
lcd.print("*C");

break;
}
if(ir > FINGER_ON){
Temperature = mlx.readObjectTempC();
lcd.setCursor(0,0);
lcd.print("Oxygen % = ");
lcd.setCursor(11,0);
lcd.print(ESpO2);
lcd.print(" ");
lcd.print("% ");
// Temperature = Temperature+2;
lcd.setCursor(0,1);
lcd.print("Temperature: ");
```

Algorithm/Code

```
      lcd.print(Temperature);
      lcd.print(" *C");
      if((ESpO2 >= 90) && (Temperature < 38)){
        digitalWrite(Redled,LOW);
        digitalWrite(Greenled,HIGH);
      }
      if((ESpO2 < 90) || (Temperature > 37)){
        digitalWrite(Greenled,LOW);
        digitalWrite(Redled,HIGH);

      }

      }

    }
  }
  if ((i % Num) == 0) {
    double R = (sqrt(sumredrms) / avered) / (sqrt(sumirrms) / aveir);
    // Serial.println(R);
    SpO2 = -23.3 * (R - 0.4) + 100; //http://ww1.microchip.com/downloads/jp/AppNotes/00001525B_JP.pdf
    ESpO2 = FSpO2 * ESpO2 + (1.0 - FSpO2) * SpO2;//low pass filter
    //  Serial.print(SpO2);Serial.print(",");Serial.println(ESpO2);
    sumredrms = 0.0; sumirrms = 0.0; i = 0;
    break;
  }
  particleSensor.nextSample(); //We're finished with this sample so move to next sample
 // Serial.println(SpO2);
```

Results


Oxygen % = nan %
Temperature: 33 *C
Last test: nan %
Last temp: 33 *C
JHD 204A


No finger?
SPO2
99
SPO2
99
0
100

Conclusion

- The body closely monitors blood oxygen levels to keep them within a specific range, so that there is enough oxygen for the needs of every cell in the body. A person's blood oxygen level is an indicator of how well the body distributes oxygen from the lungs to the **cells**, and it can be important for people's health.
- Spo2 is one of the most vital components for our existence. And keeping track of it could make a difference between being healthy and well, not being there at all.
- Pulse oximetry is truly vital to medical care, but this prominence carries a responsibility. Too often, pulse oximetry readings are misinterpreted or are erroneous for various reasons. The tendency to accept the percentage at face value is high. With future technology, pulse oximeters will continue to improve patient care more effectively and efficiently.

References

- https://iopscience.iop.org/article/10.1088/1742-6596/1402/3/033110/meta

- **Measurement device for detecting oxygen saturation in blood**
  ...*iopscience.iop.org › article › meta*

- **(PDF) Design a system of measurement of heart rate, oxygen**
  ...*www.researchgate.net › publication › 328183393_Design...*